

Technical Profile

Technical Stack: Python, SQL, PyTorch, JAX, PySpark, Databricks, Inspect, Logfire, wandb, git, LaTeX

Research specializations: LLM Post-training, Supervised Fine-Tuning, Reinforcement Learning, LLM agents, LLM evaluations, AI alignment, Scalable Oversight, OCR

Experience

Research Scholar - MATS (ML Alignment and Theory Scholars)

January 2026 – Current, Berkeley, CA (On-site)

- Doing ELK-type research by setting competing hypotheses and designing experiments to discriminate between them based on behavioral signatures each hypothesis would predict differently.
- Developing automated model behaviour investigation pipeline using steering vectors and linear probes
- Working on novel research agenda with Prof. Shi Feng

Research Engineer - Palisade Research

September 2024 – November 2025, Berkeley, CA (Remote)

- Authored research paper demonstrating autonomous LLM agent solutions for cybersecurity challenges: "Hacking CTFs with Plain Agents" (arXiv:2412.02776).
- Evaluated real-time voice agents on the possibility of autonomously passing technical interviews. Link: <https://github.com/PalisadeResearch/pass-interview>
- Trained and evaluated LLM agents using supervised fine-tuning and reinforcement learning on single-machine and multi-node network attack environments. Link: <https://github.com/PalisadeResearch/hackingdemo>
- Organized Misalignment Bounty event, evaluating 300+ cases to produce public dataset and analysis report. Link: <https://huggingface.co/datasets/palisdereasearch/Misalignment-Bounty-Submissions>

Research Scholar - MATS (ML Alignment and Theory Scholars)

January 2025 – March 2025, Berkeley, CA (Remote) (concurrent with UK AISI)

- Implemented LoRA-based adversarial training methods on Llama-3, Gemma-2, and Qwen2.5-Coder models (7B-9B parameters) to improve robustness against jailbreak attempts while preserving general capabilities.

Research Collaborator - UK AI Security Institute

February 2025 – March 2025, London, UK (Remote) (concurrent with MATS)

- During "Bounty Programme" I received a grant to develop agent scaffolding for cybersecurity benchmarks using Inspect AI framework to understand the capabilities of frontier models.

Data Engineer - Wood Mackenzie

June 2023 – December 2024, Astana, Kazakhstan

- Built data extraction pipeline on Databricks platform, processing 100+ GB of records daily.
- Optimized data cleaning algorithm through numerical method testing, increasing usable dataset coverage from 88% to 95%.
- Implemented fine-tuning pipeline for Vision Transformer-based OCR systems, improving data completeness by 10%.

Research Assistant - Department of Mechanical and Aerospace Engineering, Nazarbayev University

May 2021 – September 2021, Astana, Kazakhstan

- Developed Physics-Informed Neural Network (PINN) model to predict blood flow rate and pressure at different points within arteries.

Research Assistant - NPC Lab, Nazarbayev University

August 2020 – December 2020, Astana, Kazakhstan

- Implemented DCGAN to generate synthetic pneumonia chest X-ray dataset to augment training data for classification models.

Education

Bachelor of Science in Computer Science, Minor in Mathematics

- Nazarbayev University, August 2018 – June 2023
- Relevant coursework: GPU Programming, Compiler Construction, Deep Learning, Advanced Algorithms and Data Structures, Formal Languages, Abstract Algebra, Advanced Linear Algebra, Statistics

Publications

- "Hacking CTFs with Plain Agents" - arXiv:2412.02776 (2024) - First Author
- "Building Comparative Motivation Profiles with Instrumental Interventions" - arXiv:2606.08243 (2026) - First Author
- "CayleyPy RL: Pathfinding and Reinforcement Learning on Cayley Graphs" - arXiv:2502.18663 (2025) - Co-Author **Accepted at Advances in Theoretical and Mathematical Physics Volume 30 (2026) Number 1.**
- "CayleyPy Growth: Efficient growth computations and hundreds of new conjectures on Cayley graphs (Brief version)" - arXiv:2509.19162 (2025) - Co-Author. **Accepted with contributed talk at NeurIPS MATH-AI Workshop.**
- "Misalignment Bounty: Crowdsourcing AI Agent Misbehavior" - arXiv:2510.19738 (2025) - First Author